

ANCA Small-Vessel Vasculitis — GPA, MPA, EGPA



1. Pauci-immune crescentic GN

WHAT YOU SEE

Pauci-immune banner over the wolf

WHAT IT ENCODES

All three ANCA vasculitides cause pauci-immune crescentic GN: rapidly progressive GN with crescents but LITTLE/NO immune deposits on IF or EM. This is the unifying renal lesion. Necrotizing, segmental glomerular injury with breaks in GBM.

BOARD TRAP

Immune-complex GN (lupus, IgA, PIGN) shows BRIGHT granular IF — pauci-immune = scant/negative IF is the discriminator.

2. GPA (Wegener's)

WHAT YOU SEE

Left workbench labeled GPA — Wegener's

WHAT IT ENCODES

Granulomatosis with polyangiitis: granulomas + necrotizing vasculitis of upper (sinusitis, saddle-nose, otitis) AND lower respiratory tract (cavitary nodules, hemoptysis) PLUS GN. c-ANCA / PR3-ANCA positive. Treat with steroids + rituximab or cyclophosphamide.

BOARD TRAP

Saddle-nose and lung CAVITATIONS point to GPA, not MPA — MPA has no granulomas and no destructive ENT disease.

3. PR3-ANCA / c-ANCA

WHAT YOU SEE

PR3 / c-ANCA tag on the GPA figure

WHAT IT ENCODES

GPA is classically c-ANCA with PR3 specificity. PR3-ANCA associates with higher relapse rate (~50%). Cytoplasmic staining pattern on immunofluorescence of neutrophils.

BOARD TRAP

Don't lock c-ANCA to lungs alone — PR3 can occur in MPA and renal-limited disease; the antigen, not just the pattern, is what's tested.

4. MPA (microscopic polyangiitis)

WHAT YOU SEE

Middle bench labeled MPA

WHAT IT ENCODES

Microscopic polyangiitis: necrotizing small-vessel vasculitis WITHOUT granulomas and WITHOUT asthma/eosinophilia. Pulmonary-renal syndrome (alveolar hemorrhage + GN) is common. p-ANCA / MPO-ANCA positive.

BOARD TRAP

MPA vs GPA: both can hit lung+kidney, but absence of granulomas and absence of ENT/sinus destruction favors MPA.

5. MPO-ANCA / p-ANCA

WHAT YOU SEE

MPO / p-ANCA tag on MPA figure

WHAT IT ENCODES

MPA and EGPA are typically p-ANCA with MPO specificity (perinuclear pattern). MPO-ANCA disease tends to be more chronic/fibrotic at diagnosis with lower relapse than PR3.

BOARD TRAP

Drug-induced ANCA (hydralazine, propylthiouracil, levamisole-cut cocaine) is classically high-titer MPO/p-ANCA — check the med list before calling primary MPA.

6. EGPA (Churg-Strauss)

WHAT YOU SEE

Right bench labeled EGPA — Churg-Strauss

WHAT IT ENCODES

Eosinophilic granulomatosis with polyangiitis: asthma + allergic rhinitis + peripheral eosinophilia, then vasculitic phase. Mononeuritis multiplex, cardiac and GI involvement; GN is less frequent. ANCA positive in only ~40% (MPO when present).

BOARD TRAP

Renal disease is LEAST common in EGPA — a question pairing asthma + eosinophilia + foot-drop (mononeuritis) is EGPA, not GPA.

7. Asthma + allergy + eosinophilia

WHAT YOU SEE

Flowers/eosinophil cloud at EGPA bench

WHAT IT ENCODES

The EGPA prodrome: late-onset asthma and allergic rhinitis with marked peripheral eosinophilia (>10%) precede vasculitis by years. Elevated IgE. Mepolizumab (anti-IL-5) is approved for relapsing/refractory EGPA.

BOARD TRAP

Eosinophilia + asthma can mimic eosinophilic pneumonia or ABPA — vasculitic features (purpura, neuropathy, GN) distinguish EGPA.

8. Mononeuritis multiplex

WHAT YOU SEE

Mononeuritis label at EGPA

WHAT IT ENCODES

Asymmetric involvement of named peripheral nerves (e.g., foot drop + wrist drop) from vasculitic ischemia of vasa nervorum. Hallmark of EGPA and PAN; can occur in any ANCA vasculitis.

BOARD TRAP

Symmetric stocking-glove neuropathy is NOT mononeuritis multiplex — the asymmetry/named-nerve pattern is the discriminator.

9. Hemoptysis / alveolar hemorrhage

WHAT YOU SEE

Hemoptysis label, upper left

WHAT IT ENCODES

Diffuse alveolar hemorrhage produces hemoptysis, dropping hemoglobin, and a rising DLCO. Pulmonary-renal syndrome (alveolar hemorrhage + GN) is the medical emergency driving urgent immunosuppression and plasmapheresis.

BOARD TRAP

Pulmonary-renal syndrome differential also includes anti-GBM (Goodpasture) and lupus — ANCA serology and IF pattern separate them.

10. High mortality — urgent Tx

WHAT YOU SEE

Red 'high mortality untreated — urgent Tx' banner

WHAT IT ENCODES

Untreated ANCA vasculitis with RPGN has high mortality; treat empirically while serologies pend if clinical suspicion is high. Induction = high-dose steroids + rituximab or cyclophosphamide. Add plasmapheresis for severe AKI/dialysis or DAH (PEXIVAS-guided).

BOARD TRAP

Waiting for ANCA/biopsy before treating RPGN loses nephrons — board answer is start immunosuppression empirically in fulminant disease.

11. Induction → maintenance

WHAT YOU SEE

Induction shelf; maintenance 1–2 yrs

WHAT IT ENCODES

Induction (3–6 mo): steroids + rituximab (or cyclophosphamide). Maintenance (≥ 1 –2 yrs): rituximab or azathioprine to prevent relapse. Rituximab is now preferred for both phases, especially PR3 disease.

BOARD TRAP

Stopping immunosuppression at remission causes relapse — maintenance therapy is required, longest in PR3/GPA.

12. Plasmapheresis

WHAT YOU SEE

Plasmapheresis label near truck

WHAT IT ENCODES

Plasma exchange removes circulating ANCA; per PEXIVAS, reserve for severe renal failure (Cr high/dialysis-dependent) or diffuse alveolar hemorrhage rather than routine use. Avacopan (C5a receptor blocker) is a steroid-sparing adjunct.

BOARD TRAP

Plasmapheresis is mandatory in anti-GBM but selective in ANCA vasculitis — don't reflexively apply Goodpasture rules to ANCA.
